CBSE Class 10 Maths (Basic) Question Paper 2022

Set – 1(430/4/1) Solutions

Section A

From Question Numbers 1 to 6 carry 2 marks each.

1. Find the value of 'k' so that the quadratic equation $3x^2 - 5x - 2k = 0$ has real and equal roots. 2

Ans:
For the quadratic equation to have real and equal roots, the discriminant (Δ) must be zero.

$\Delta = b^2 - 4ac$

Here, a = 3, b = −5, and c = −2k.

$(-5)^2 - 4(3)(-2k) = 0$

25+12(2k) = 0

25 + 24k= 0

24k = −25

$k = -\frac{25}{24}$

Therefore, $k = -\frac{25}{24}$

2. (a) Determine the 36th term of the A.P. whose first two terms are -3 and 4 respectively.

OR

(b) Write the next two terms of the A.P.: $\sqrt{27}, \sqrt{48}, \sqrt{75}, \ldots$ **2**

Ans:

(a) First term a = −3

Second term $a_2 = 4$

Common difference d = a_2 - a = 4 - (-3) = 7

The nth term of an A.P. is given by:

a_n = a + (n-1) d

a_{36} = -3 + (36-1) · 7

a_{36} = -3 + 35 · 7

a_{36} = -3 + 245

$a_{36} = 242

Therefore, the 36th term is 242.

OR

(b) The given sequence is $\sqrt{27}, \sqrt{48}, \sqrt{75}, \ldots$

These can be simplified to:

$\sqrt{27} = 3\sqrt{3}, \quad \sqrt{48} = 4\sqrt{3}, \quad \sqrt{75} = 5\sqrt{3}$

The sequence in simplified form is:

$3\sqrt{3}, 4\sqrt{3}, 5\sqrt{3}, \ldots$

The common difference is d = $4\sqrt{3} - 3\sqrt{3} = \sqrt{3}$

Next term after $5\sqrt{3}$:

$5\sqrt{3} + \sqrt{3} = 6\sqrt{3}$

Next term after $6\sqrt{3}$:

$6\sqrt{3} + \sqrt{3} = 7\sqrt{3}$

Therefore, the next two terms are $6\sqrt{3}$ and $7\sqrt{3}$.

3. Using the empirical relationship between the three measures of central tendency, find the median of a distribution, whose mean is 169 and mode is 175. 2

Ans: The empirical relationship between mean, median, and mode is given by:

Mean – Mode = 3(Mean−Median)

Given, Mean = 169 and Mode = 175:

169 – 175 = 3(169 − Median)

−6 = 3(169 − Median)

−6 = 507 − 3Median

3Median = 507 + 6

3Median = 513

Median = $\frac{513}{3}$ = 171

Therefore, the median is 171.

4. The frequency distribution table of agriculture holding in a village is given below:

Area of Land (in hectares)	1 – 3	3 – 5	5 – 7	7 – 9	9 – 11	11 – 13
13Number of families	20	45	80	55	40	12

Find the modal agriculture holding per family. 2

Ans: The mode is the class with the highest frequency.

Here, the highest frequency is 80 in the class interval 5 – 7.

Therefore, the modal agriculture holding per family is 5 – 7 hectares.

5. The distance between two tangents parallel to each other of a circle is 13 cm. Find the radius of the circle. 2

Ans: Let the radius of the circle be r.

The distance between two parallel tangents of a circle is equal to the diameter of the circle.

Therefore, the diameter of the circle is 13 cm: 2r = 13

$r = \frac{13}{2} = 6.5$ cm

Therefore, the radius of the circle is 6.5 cm.

6. (a) Find the curved surface area of a right circular cone whose height is 15 cm and base radius is 8 cm. [Use $\pi = \frac{22}{7}$] 2

OR

(b) The surface area of a sphere is 616 sq cm. Find its radius. [Use $\pi = \frac{22}{7}$]

Ans:

(a) The curved surface area of a cone is given by:

CSA = πrl

where r is the base radius and l is the slant height.

The slant height l can be found using Pythagoras theorem:

$$l = \sqrt{r^2 + h^2} = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17\,\text{cm}$$

$$\text{CSA} = \pi rl = \pi rl = \frac{22}{7} \times 8 \times 17 = \frac{22 \times 8 \times 17}{7} = \frac{2992}{7} \approx 427.43 \text{ sq cm}$$

Therefore, the curved surface area of the cone is approximately 427.43 sq cm.

OR

(b) The surface area of a sphere is given by: $4\pi r^2$

Given, the surface area is 616 sq cm:

$$4\pi r^2 = 616$$

$$4 \times \frac{22}{7} \times r^2 = 616$$

$$r^2 = \frac{616 \times 7}{4 \times 22} = \frac{4312}{88} = 49$$

r = $\sqrt{49}$ = 7 cm

Therefore, the radius of the sphere is 7 cm.

Section B

Question Numbers 7 to 10 carry 3 marks each

7. (a) If x = 3 is one root of the quadratic equation $2x^2 + px + 30 = 0$, find the value of p and the other root of the quadratic equation. 3

OR

(b) The length of a rectangular park is 5 metres more than twice its breadth. If the area of the park is 250 sq m, find the length and breadth of the park.

Ans:

(a) Given x = 3 is one root of the quadratic equation $2x^2 + px + 30 = 0$

Substitute x = 3 into the equation:

$2(3)^2 + 3p + 30 = 0$

2(9) + 3p + 30 = 0

18+3p+30=018 + 3p + 30 = 018+3p+30=0 3p=−483p = -483p=−48 p=−16p = -16p=−16

To find the other root, use the sum and product of the roots:

Sum of the roots = $-\frac{b}{a}$ = $-\frac{p}{2}$ $\Rightarrow$ −ab = −2p

Sum of roots = 3 + r = $-\frac{-16}{2}$ = 8

r = 8 – 3 = 5

Therefore, the other root is 5 and p = −16

OR

(b) Let the breadth of the park be x metres.

The length of the park is 2x + 5 metres.

Given, the area of the park is 250 sq m:

x (2x + 5) = 250

$2x^2 + 5x - 250 = 0$

Solving this quadratic equation using the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where a = 2, b = 5, and c = −250.

$$x = \frac{-5 \pm \sqrt{(5)^2 - 4(2)(-250)}}{4} = \frac{-5 \pm \sqrt{25 + 2000}}{4} = \frac{-5 \pm \sqrt{2025}}{4} = \frac{-5 \pm 45}{4}$$

Therefore,

$x = \frac{40}{4} = 10$ (breadth)

$x = \frac{-50}{4}$ (not possible as length cannot be negative)

The length of the park:

2x + 5 = 2(10) + 5 = 25 metres

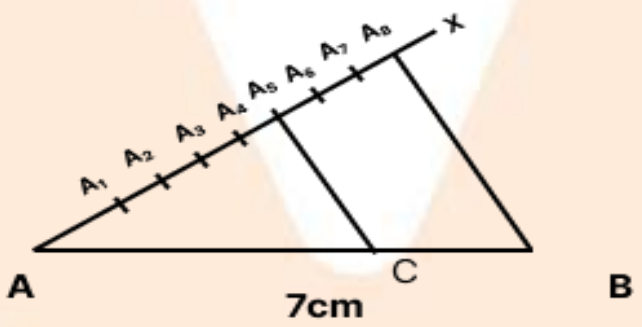
Therefore, the length is 25 metres and the breadth is 10 metres.

8. Draw a line segment of length 7 cm and divide it in the ratio 5 : 3. **3**

Ans:
Follow the steps below to divide the line segment in the ratio 5:3:

- Draw a line segment AB of length 7 cm.
- Draw a ray AX making an acute angle with AB.
- Mark 8 equal divisions on AX (since 5 + 3 = 8) and label them as $A_1, A_2, \ldots A_8$
- Join B to A_8.
- Draw a line through A_5 (5th division) parallel to A_8B, intersecting AB at point C.

Point C divides AB in the ratio 5:3.



9. Find the sum of first 16 terms of the A.P. whose n^{th} term is given by a_n = 5n - 3. 3

Ans: The nth term of the A.P. is a_n = 5n - 3.

The first term a = a_1 = 5(1) - 3 = 2

The common difference d = $a_2 - a_1 = (5 \cdot 2 - 3)$ - $(5 \cdot 1 - 3)$ = 7 - 2 = 5

The sum of the first n terms of an A.P. is given by:

$$S_n = \frac{n}{2}\left(2a + (n-1)d\right)$$

For the first 16 terms:

$$S_{16} = \frac{16}{2}\left(2 \cdot 2 + (16-1) \cdot 5\right)$$

$$S_{16} = 8\left(4 + 75\right) S16? = 8(4+75) S16 = 8 \times 79 = 632$$

Therefore, the sum of the first 16 terms is 632.

10. Two poles of heights 25 m and 35 m stand vertically on the ground. The tops of two poles are connected by a wire, which is inclined to the horizontal at an angle of 30°. Find the length of the wire and the distance between the poles. 3

Ans: Let the distance between the two poles be d metres and the length of the wire be L.

The height difference between the poles is 35−25=1035 - 25 = 1035−25=10 metres.

Using trigonometry,

$$\tan 30^\circ = \frac{\text{height difference}}{\text{distance between poles}} = \frac{10}{d}$$

$$\frac{1}{\sqrt{3}} = \frac{10}{d}$$

d = $10\sqrt{3} \approx 17.32$ metres

To find the length of the wire L:

Using the Pythagorean theorem in the right triangle formed,

L = $\sqrt{d^2 + (\text{height difference})^2}$

L = $\sqrt{(10\sqrt{3})^2 + 10^2}$

L = $\sqrt{300 + 100}$

L = $\sqrt{400}$ = 20 metres

Therefore, the length of the wire is 20 metres and the distance between the poles is approximately 17.32 metres.

Section C

Question Numbers 11 to 14 carry 4 marks each.

11. As observed from the top of a 100 m high lighthouse from the sea-level, the angles of depression of two ships are found to be 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships. Use $\sqrt{3} = 1.732$

4

Ans: Let the distance of the closer ship from the base of the lighthouse be d_1 and the distance of the farther ship be d_2.

Using trigonometry,

For the ship at an angle of depression 45°:

$$\tan 45^\circ = \frac{100}{d_1} = 1$$

$$d_1 = 100 \text{ metres}$$

For the ship at an angle of depression 30°:

$$\tan 30^\circ = \frac{100}{d_2} = \frac{1}{\sqrt{3}}$$

$$d_2 = 100\sqrt{3} = 100 \times 1.732 = 173.2 \text{ metres}$$

The distance between the two ships is:

$$d_2 - d_1 = 173.2 - 100 = 73.2 \text{ metres}$$

Therefore, the distance between the two ships is 73.2 metres.

12. (a) Prove that the opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle. 4

OR

(b) If a circle is touching the side BC of △ABC at P and is touching AB and AC produced at Q and R respectively (see the figure)

Prove that AQ = $\frac{1}{2}$ (perimeter of △ABC).

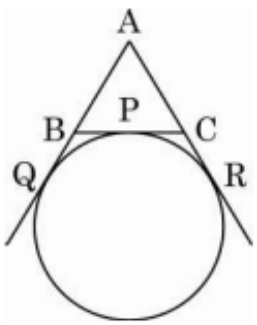


Fig. – 1

Ans:

(a) Let the quadrilateral be ABCD circumscribing a circle with center O.

Let the circle touch the sides AB, BC, CD, and DA at points P, Q, R, and S respectively.

Since the tangents drawn from an external point to a circle are equal, we have:

AP = AS, BP = BQ, CR = CQ, DR = DS

Adding these,

AP + BP + CR + DR = AS + BQ + CQ + DS

Simplifying,

AB + CD = AD + BC

Since the sum of opposite sides are equal, the opposite sides of the quadrilateral subtend supplementary angles at the center of the circle.

OR

(b) Given: A circle touches the side BC of △ABC at P and touches AB and AC produced at Q and R respectively.

To prove: AQ = $\frac{1}{2}$ (perimeter of △ABC)

Let AB = c, BC = a, and CA = b.

The tangents drawn from an external point to a circle are equal.

So,

AQ = AR, BQ = BP, CR = CP

Therefore,

AQ + BQ + CR = AR + BP + CP

Since the tangents from A to the points Q and R, B to the points P and Q, and C to the points P and R are equal in length:

AQ = AR, BQ = BP, CR = CP

Therefore,

$$AQ = \frac{a+b+c}{2} = \frac{\text{Perimeter of } \triangle ABC}{2}$$

Hence proved.

Case Study - 1

13. Electric buses are becoming popular nowadays. These buses have the electricity stored in a battery. Electric buses have a range of approximately 280 km with just one charge. These buses are superior to diesel buses as they reduce brake wear and also reduce pollution. Transport department of a city wants to buy some electric buses for the city. So, the department wants to know the distance travelled by existing public transport buses in a day.

The following data shows the distance travelled by 50 existing public transport buses in a day.

Daily distance travelled (in km)	100 – 120	120 – 140	140 – 160	160 – 180	180 – 200
Number of buses	12	14	8	6	10

(a) Find the 'median' distance travelled by a bus. **2**

(b) Find the 'mean' average distance travelled by a bus. **2**

Ans:

(a) To find the median, we first find the cumulative frequency:

Daily distance travelled (in km)	Number of buses	Cumulative frequency
100 – 120	12	12
120 – 140	14	26
140 – 160	8	34
160 – 180	6	40
180 – 200	10	50

The median class is the class interval where the cumulative frequency is greater than or equal to 25 (since $\frac{n}{2}=\frac{50}{2}=25$)

Median class is 120 – 140

The formula for median is:

$$\text{Median} = L + \frac{\frac{n}{2}-CF}{f} \cdot h$$

where L = 120, n = 50, CF = 12, f = 14, and h = 20.

Substitute these values in formula

$$\text{Median} = 120 + \frac{25-12}{14} \cdot 20$$

$$\text{Median} = 120 + \frac{13}{14} \cdot 20$$

$$\text{Median} = 120 + 18.57 \approx 138.57 \text{ km}$$

Therefore, the median distance travelled by a bus is approximately 138.57 km.

(b) First calculate the x_i for each interval

We know that $x_i = \frac{\text{lower limit} + \text{upper limit}}{2}$

For 100–120: $\frac{100+120}{2} = 110$

For 120–140: $\frac{120+140}{2} = 130$

For 140–160: $\frac{140+160}{2} = 150$

For 160–180: $\frac{160+180}{2} = 170$

For 180–200: $\frac{180+200}{2} = 190$

Now, we multiply the x_i by the frequency f_i for each class interval to find $f_i x_i$:

Daily distance travelled (in km)	**Number of buses (f_i)**	x_i	$f_i x_i$
100 – 120	12	110	1320
120 – 140	14	130	1820
140 – 160	8	150	1200
160 – 180	6	170	1020
180 – 200	10	190	1900

Next, we sum up the $f_i x_i$ values and the frequencies:

$\sum f_i = 12 + 14 + 8 + 6 + 10 = 50$

$\sum f_i x_i = 1320 + 1820 + 1200 + 1020 + 1900 = 7260$

Finally, the mean average distance travelled by a bus is calculated using the formula:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{7260}{50} = 145.2$$

Therefore, the mean average distance travelled by a bus is 145.2 km.

Case Study - 2

14. A company deals in casting and moulding of metal on orders received from its clients.

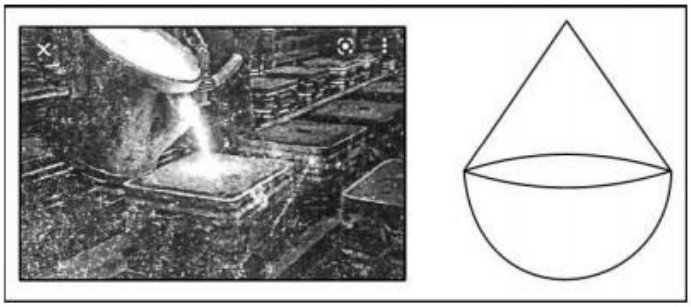


Fig. – 2

In one such order, company is supposed to make 50 toys in the form of a hemisphere surmounted by aright circular cone of the same base radius as that of hemisphere. If the radius of the base of the cone is 21 cm and height is 28 cm,

(a) find the volume of 50 toys; **2**

(b) find the ratio of the volume of hemisphere to the volume of cone. **2**

Ans:

(a) The volume of one toy is the sum of the volume of the hemisphere and the cone.

Volume of hemisphere:

$$V_h = \frac{2}{3}\pi r^3$$

where r = 21 cm.

$$V_h = \frac{2}{3}\pi(21)^3 = \frac{2}{3}\pi \times 9261 = \frac{2\times 9261}{3}\pi = 6174\pi \text{cm}^3$$

Volume of cone:

$$V_c = \frac{1}{3}\pi r^2$$

where r = 21 cm.

$$V_c = \frac{1}{3}\pi(21)^2 \times 28 = \frac{1}{3}\pi \times 441 \times 28 = \frac{441 \times 28}{3}\pi = 4116\pi \text{cm}^3$$

Total volume of one toy:

$$V = V_h + V_c = 6174\pi + 4116\pi = 10290\pi \text{cm}^3$$

Volume of 50 toys:

$$V_{50} = 50 \times 10290\pi = 514500\pi \text{cm}^3$$

Therefore, the volume of 50 toys is $514500\pi \text{cm}^3$

(b) Ratio of the volume of the hemisphere to the volume of the cone:

$$\frac{V_h}{V_c} = \frac{6174\pi}{4116\pi} = \frac{6174}{4116} = \frac{1029}{686}$$

Therefore, the ratio of the volume of the hemisphere to the volume of the cone is $\frac{1029}{686}$